

Assignment_1:User analytics based on sample data

```
import pandas as pd
import numpy as np

from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.linear_model import LinearRegression

from sklearn.model_selection import train_test_split

from sklearn.metrics import accuracy_score, confusion_matrix

from sklearn.preprocessing import LabelEncoder
```

```
df = pd.read_csv("Dataset 2.csv")
print(df.head())
```

	UserID	Age	Gender	SubscriptionType	WatchHoursPerWeek	DevicesUsed	\
0	1001	22	Female	Basic	23	5	
1	1002	55	Male	Basic	9	4	
2	1003	49	Male	Basic	8	3	
3	1004	39	Female	Premium	19	5	
4	1005	38	Female	Premium	23	5	

	FavoriteGenre	AdClicks	MonthlySpend	SubscriptionRenewed
0	Comedy	13	353	No
1	Drama	14	317	Yes
2	Comedy	16	309	No
3	Drama	45	833	Yes
4	Sci-Fi	24	804	Yes

```
print("Number of rows and columns in dataset:",df.shape)
```

```
Number of rows and columns in dataset: (750, 10)
```

```
print("Column names:", df.columns)
```

```
Column names: Index(['UserID', 'Age', 'Gender', 'SubscriptionType', 'WatchHoursPerWeek',
                     'DevicesUsed', 'FavoriteGenre', 'AdClicks', 'MonthlySpend',
                     'SubscriptionRenewed'],
                     dtype='object')
```

```
print("Numerical and categorical values:\n",df.dtypes)
```

```
Numerical and categorical values:
UserID          int64
Age             int64
Gender          object
SubscriptionType object
WatchHoursPerWeek int64
DevicesUsed     int64
FavoriteGenre   object
AdClicks        int64
MonthlySpend    int64
SubscriptionRenewed object
dtype: object
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 750 entries, 0 to 749
Data columns (total 10 columns):
#   Column              Non-Null Count  Dtype
---  -
0   UserID              750 non-null   int64
1   Age                 750 non-null   int64
2   Gender              750 non-null   object
3   SubscriptionType     750 non-null   object
4   WatchHoursPerWeek    750 non-null   int64
5   DevicesUsed         750 non-null   int64
6   FavoriteGenre        750 non-null   object
7   AdClicks            750 non-null   int64
8   MonthlySpend        750 non-null   int64
9   SubscriptionRenewed  750 non-null   object
dtypes: int64(6), object(4)
memory usage: 58.7+ KB
```

```
print("Average age of users:", df["Age"].mean())
```

Average age of users: 41.824

```
print("Average watch hours per week:", df["WatchHoursPerWeek"].mean())
```

Average watch hours per week: 14.236

```
print("Average monthly spending of users:", round(df["MonthlySpend"].mean(),2))
```

Average monthly spending of users: 689.91

```
print("Number of users in each subscription category:\n",df.SubscriptionType.value_counts())
```

Number of users in each subscription category:
SubscriptionType
Basic 342
Premium 279
VIP 129
Name: count, dtype: int64

```
print("Percentage of users who renewed their subscription:",round((df['SubscriptionRenewed']=='Yes').mean()*100,2))
```

Percentage of users who renewed their subscription: 46.27

```
le = LabelEncoder()  
df["Gender"]=le.fit_transform(df['Gender'])  
df["SubscriptionType"] = le.fit_transform(df['SubscriptionType'])  
df["FavoriteGenre"] = le.fit_transform(df['FavoriteGenre'])
```

```
x = df.drop(["UserID", "SubscriptionRenewed"],axis=1)  
y = df["SubscriptionRenewed"]  
  
x_train, x_test, y_train, y_test = train_test_split(  
    x,y,test_size=0.2,random_state=42  
)  
  
dt=DecisionTreeClassifier(random_state=42)  
dt.fit(x_train,y_train)  
pred = dt.predict(x_test)  
  
print("Decision tree accuracy:", accuracy_score(y_test,pred))  
print("Confusion matrix:\n", confusion_matrix(y_test,pred))
```

Decision tree accuracy: 0.56
Confusion matrix:
[[48 34]
 [32 36]]

```
knn = KNeighborsClassifier(n_neighbors=5)  
knn.fit(x_train,y_train)  
knn_pred= knn.predict(x_test)  
print("KNN accuracy:",accuracy_score(y_test,knn_pred))
```

KNN accuracy: 0.6333333333333333

```
x_reg=df.drop(["UserID", "SubscriptionRenewed"],axis=1)  
y_reg=df["MonthlySpend"]  
  
x_train_reg,x_test_reg,y_train_reg,y_test_reg=train_test_split(  
    x_reg,y_reg,test_size=0.2,random_state=42  
)  
  
lr=LinearRegression()  
lr.fit(x_train_reg,y_train_reg)  
amount = lr.predict([[25,0,0,15,3,1,12,220]])  
print("Predicting Monthly spending:", amount[0])
```

Predicting Monthly spending: 220.00000000000026
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, warnings.warn(

